

Data Mining SENG 474 Assignment 1

Part 1: Complete the attached notebook [100 marks]

The attached jupyter notebook has multiple components and they are as follows:

1. Perform an initial exploration of the data:
 - a. Looking for missing data
 - b. Visualizing the data
 - c. Examining correlations between features
 - d. Analyzing the distribution the data is drawn from
 - e. Performing what you feel to be relevant preprocessing of the data such as imputation, feature transformations (log or polynomial transforms), removing redundant features or outliers from the data, etc.
2. Preparing training, validation, and test sets for each model (if you choose to implement/use K-Fold cross validation you do not need to create an explicit validation set):
 - a. Splitting the dataset into parts
 - b. Normalizing the features
3. Implementing *from scratch* at least 1 of the following:
 - a. Linear Regression
 - b. Logistic Regression
 - c. K-Means Clustering
 - d. K-Fold Cross Validation
4. Implementing *from scratch* at least 1 of the following for either classification or regression but not both):
 - a. Support Vector Machine
 - b. Neural Network (Multi-Layer Perceptron)
 - c. Decision Trees
5. Implementing *using sklearn* 2 other models such that you have implemented at least
 - a. 1 regression model

- b. 1 classification model
 - c. 1 unsupervised learning model (K-Means Clustering)
 - d. 1 additional regression or classification model
6. Evaluate your models and tune their hyperparameters using the validation set.
 - a. For each model you will need to perform 2 experiments in which you vary a single hyper-parameter and keep all others fixed.
 - b. In total this will mean that you conduct 8 experiments.
7. Finally, you will evaluate your models on the test set and compare their performances.
 - a. This does not apply to K-Means clustering as there is no test set there.

Part 2: Write a short report (2-page min, 4-page max) about your work in LaTeX (I recommend using overleaf) [100 marks]

The report will generally serve as a reflection about what you did and:

1. Clearly indicate the models/algorithms you chose to implement and brief explanation of why you chose them.
2. Discuss any insight gained from your initial data exploration.
3. Provide justification for any pre-processing steps that you took.
4. Include a performance analysis of each model discussing the impact of the hyper-parameter tuning that was conducted and the effects that had on the models in terms of overfitting and underfitting the data.
5. Include a comparative analysis of the two models that shared an objective using the performance of the models on the test set in your analysis.
6. Conclude with a summary of your findings.

Other things to keep in mind:

- Don't touch the training set until the very end.
- Plot error curves not accuracy curves.
- Frame your report's discussion as lowering error not increasing accuracy, it will make it easier for the TAs if everyone just does this the same way.

- When conducting multiple experiments involving hyperparameter tuning you can just use a for-loop to iterate through various hyperparameters.
- A good experiment involves keeping all, but one hyperparameter fixed, and then varying one hyperparameter along the x-axis and plotting *both* training error and validation or test error (ideally, with both curves in the same plot).
- For each method, try to experiment with at least two hyperparameters (one of which can be the size of the training set).

Grading:

Part	1.1	1.2	1.3	1.4	1.5	1.6	1.7	2.1	2.2	2.3	2.4	2.5
Points	10	5	15	25	15	15	15	5	10	5	50	30

If you are struggling to implement a model from scratch, I advise you to come to office hours for assistance.

Submission:

I recommend you use Google-Colab for your notebooks as it makes troubleshooting any package issues much easier.

Before you submit make sure to Restart Kernel and Run All Cells to ensure your notebook work as intended.

Please provide any links to your notebook (if using Google-Colab ensure sharing permissions are properly set) in a .txt file with a short description, a single .pdf of all notebook(s), a .pdf version of your report, and any .py or .ipynb files you used. Code files may be uploaded as a .zip file but please ensure you upload the .pdf files separately as it makes grading easier for the TAs.